Logistic Regression – Airline Customer Satisfaction Prediction

Project Overview

This project applies Logistic Regression to predict whether an airline passenger is satisfied based on demographic information, travel characteristics, and service ratings.

The analysis includes data inspection, preprocessing, categorical variable encoding, model development, model evaluation using classification metrics, interpretation of feature importance, and business recommendations aimed at improving customer satisfaction.

```
#Import Libraries
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import (
    confusion_matrix,
    ConfusionMatrixDisplay,
    classification_report,
    accuracy_score,
    precision_score,
    recall_score
)
```

```
#Load Dataset
df = pd.read_csv("Invistico_Airline.csv")
```

Inspections

```
df.head()
```

```
df.shape
```

```
(129880, 22)
```

```
df.columns
```

```
Index(['satisfaction', 'Customer Type', 'Age', 'Type of Travel', 'Class',
       'Flight Distance', 'Seat comfort', 'Departure/Arrival time convenient',
       'Food and drink', 'Gate location', 'Inflight wifi service',
       'Inflight entertainment', 'Online support', 'Ease of Online booking',
       'On-board service', 'Leg room service', 'Baggage handling',
       'Checkin service', 'Cleanliness', 'Online boarding',
       'Departure Delay in Minutes', 'Arrival Delay in Minutes'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 129880 entries, 0 to 129879
Data columns (total 22 columns):
 #   Column                             Non-Null Count   Dtype
---  ------                             --------------   -----
 0   satisfaction                       129880 non-null  object
 1   Customer Type                      129880 non-null  object
 2   Age                                129880 non-null  int64
 3   Type of Travel                     129880 non-null  object
```

```
 4   Class                              129880 non-null  object
 5   Flight Distance                    129880 non-null  int64
 6   Seat comfort                       129880 non-null  int64
 7   Departure/Arrival time convenient  129880 non-null  int64
 8   Food and drink                     129880 non-null  int64
 9   Gate location                      129880 non-null  int64
 10  Inflight wifi service              129880 non-null  int64
 11  Inflight entertainment             129880 non-null  int64
 12  Online support                     129880 non-null  int64
 13  Ease of Online booking             129880 non-null  int64
 14  On-board service                   129880 non-null  int64
 15  Leg room service                   129880 non-null  int64
 16  Baggage handling                   129880 non-null  int64
 17  Checkin service                    129880 non-null  int64
 18  Cleanliness                        129880 non-null  int64
 19  Online boarding                    129880 non-null  int64
 20  Departure Delay in Minutes         129880 non-null  int64
 21  Arrival Delay in Minutes           129487 non-null  float64
dtypes: float64(1), int64(17), object(4)
memory usage: 21.8+ MB
```

```
df.describe(include='all')
```

Data Cleaning

The dataset was examined for missing values, duplicate records, and incorrect data types to ensure suitability for Logistic Regression modeling.

```
#Checking for missing value
df.isnull().sum()
```

```
satisfaction                          0
Customer Type                         0
Age                                   0
Type of Travel                        0
Class                                 0
Flight Distance                       0
Seat comfort                          0
Departure/Arrival time convenient     0
Food and drink                        0
Gate location                         0
Inflight wifi service                 0
Inflight entertainment                0
Online support                        0
Ease of Online booking                0
On-board service                      0
Leg room service                      0
Baggage handling                      0
Checkin service                       0
Cleanliness                           0
Online boarding                       0
Departure Delay in Minutes            0
Arrival Delay in Minutes            393
dtype: int64
```

```
#Checking for duplicate
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.dtypes
```

```
satisfaction                         object
Customer Type                        object
Age                                   int64
Type of Travel                       object
Class                                object
Flight Distance                       int64
Seat comfort                          int64
Departure/Arrival time convenient     int64
Food and drink                        int64
Gate location                         int64
Inflight wifi service                 int64
Inflight entertainment                int64
Online support                        int64
```

```
Ease of Online booking              int64
On-board service                    int64
Leg room service                    int64
Baggage handling                    int64
Checkin service                     int64
Cleanliness                         int64
Online boarding                     int64
Departure Delay in Minutes          int64
Arrival Delay in Minutes          float64
dtype: object
```

```
(df.isnull().sum()/len(df))*100
```

```
satisfaction                      0.000000
Customer Type                     0.000000
Age                               0.000000
Type of Travel                    0.000000
Class                             0.000000
Flight Distance                   0.000000
Seat comfort                      0.000000
Departure/Arrival time convenient 0.000000
Food and drink                    0.000000
Gate location                     0.000000
Inflight wifi service             0.000000
Inflight entertainment            0.000000
Online support                    0.000000
Ease of Online booking            0.000000
On-board service                  0.000000
Leg room service                  0.000000
Baggage handling                  0.000000
Checkin service                   0.000000
Cleanliness                       0.000000
Online boarding                   0.000000
Departure Delay in Minutes        0.000000
Arrival Delay in Minutes          0.302587
dtype: float64
```

```
df['Arrival Delay in Minutes'] = df['Arrival Delay in Minutes'].fillna(
    df['Arrival Delay in Minutes'].median()
)
```

```
df.isnull().sum()
```

```
satisfaction                      0
Customer Type                     0
Age                               0
Type of Travel                    0
Class                             0
Flight Distance                   0
Seat comfort                      0
Departure/Arrival time convenient 0
Food and drink                    0
Gate location                     0
Inflight wifi service             0
Inflight entertainment            0
Online support                    0
Ease of Online booking            0
On-board service                  0
Leg room service                  0
Baggage handling                  0
Checkin service                   0
Cleanliness                       0
Online boarding                   0
Departure Delay in Minutes        0
Arrival Delay in Minutes          0
dtype: int64
```

Target Variable Inspection

The target variable (satisfaction) was examined to determine the distribution of satisfied and dissatisfied passengers before model development.

```
#Count Classes
df['satisfaction'].value_counts()
```

```
satisfaction
satisfied       71087
```

```
dissatisfied    58793
Name: count, dtype: int64
```

```
#Percentage
df['satisfaction'].value_counts(normalize=True)*100
```

```
satisfaction
satisfied       54.73283
dissatisfied    45.26717
Name: proportion, dtype: float64
```

```
#Visualization
plt.figure(figsize=(6,5))

sns.countplot(
    data=df,
    x='satisfaction'
)

plt.title("Passenger Satisfaction Distribution")
plt.xlabel("Satisfaction")
plt.ylabel("Count")

plt.show()
```

Interpretation

The distribution of satisfied and dissatisfied passengers was examined before model training.

The dataset appears reasonably balanced, making it suitable for Logistic Regression without requiring additional techniques such as oversampling or undersampling.

Interpretation

The target variable shows that approximately **54.73%** of passengers were satisfied, while **45.27%** were dissatisfied.

The distribution is relatively balanced, indicating that the dataset is suitable for Logistic Regression without requiring additional techniques to address class imbalance.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the relationships between passenger characteristics, service quality ratings, and customer satisfaction before model development.

```
#Numerical Variable Correlation
#Since the dataset contains many numerical variables, a correlation heatmap is useful.
plt.figure(figsize=(15,10))

corr = df.select_dtypes(include=['number']).corr()

sns.heatmap(
    corr,
    cmap='coolwarm',
    center=0
)

plt.title("Correlation Matrix")

plt.show()
```

```
#Satisfaction by Travel Class
plt.figure(figsize=(7,5))

sns.countplot(
    data=df,
    x='Class',
```

```
    hue='satisfaction'
)

plt.title("Passenger Satisfaction by Travel Class")

plt.show()
```

```
#Satisfaction by Customer Type
plt.figure(figsize=(7,5))

sns.countplot(
    data=df,
    x='Customer Type',
    hue='satisfaction'
)

plt.title("Customer Satisfaction by Customer Type")

plt.show()
```

```
#Satisfaction by Type of Travel
plt.figure(figsize=(7,5))

sns.countplot(
    data=df,
    x='Type of Travel',
    hue='satisfaction'
)

plt.title("Customer Satisfaction by Type of Travel")

plt.show()
```

```
#Boxplot: Let's examine whether passengers who were satisfied rated the in-flight service differently.
plt.figure(figsize=(8,5))

sns.boxplot(
    data=df,
    x='satisfaction',
    y='Inflight wifi service'
)

plt.title("Inflight WiFi Rating by Satisfaction")

plt.show()
```

Interpretation

The exploratory analysis suggests that passenger satisfaction varies across travel classes, customer types, and travel purposes.

Service quality ratings also appear to differ between satisfied and dissatisfied passengers, indicating that these variables may contribute substantially to the classification model.

Feature Engineering

Logistic Regression requires numerical input variables. Therefore, all categorical variables were encoded into numerical representations prior to model development.

```
#Encode Categorical Variables
print(df.columns.tolist())
```

```
['satisfaction', 'Customer Type', 'Age', 'Type of Travel', 'Class', 'Flight Distance', 'Seat comfort', 'Departure/Arrival time c
```

```
#Satisfied = 1
#Dissatisfied = 0
# Encode target variable
df['satisfaction'] = df['satisfaction'].map({
    'satisfied':1,
    'dissatisfied':0
})

df['satisfaction'].head()
```

```
0    1
1    1
2    1
3    1
4    1
Name: satisfaction, dtype: int64
```

```
#Encode Predictor Variables: We'll use One-Hot Encoding for the remaining categorical predictors.
df_encoded = pd.get_dummies(
    df,
    columns=[
        'Customer Type',
        'Type of Travel',
        'Class'
    ],
    drop_first=True,
    dtype=int
)
```

```
df_encoded.head()
```

```
df_encoded.columns
```

```
Index(['satisfaction', 'Age', 'Flight Distance', 'Seat comfort',
       'Departure/Arrival time convenient', 'Food and drink', 'Gate location',
       'Inflight wifi service', 'Inflight entertainment', 'Online support',
       'Ease of Online booking', 'On-board service', 'Leg room service',
       'Baggage handling', 'Checkin service', 'Cleanliness', 'Online boarding',
       'Departure Delay in Minutes', 'Arrival Delay in Minutes',
       'Customer Type_disloyal Customer', 'Type of Travel_Personal Travel',
       'Class_Eco', 'Class_Eco Plus'],
      dtype='object')
```

Interpretation

The categorical variables were successfully transformed into dummy variables using one-hot encoding. One category from each variable was omitted to avoid the dummy variable trap and serve as the reference category during model interpretation.

Feature Selection

The encoded dataset was separated into predictor variables (X) and the target variable (y). The predictors include passenger demographics, travel information, and service quality ratings, while passenger satisfaction serves as the target variable.

```
# Independent variables
X = df_encoded.drop(columns='satisfaction')

# Target
y = df_encoded['satisfaction']
```

```
#Train_Test_Split
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
```

```
    stratify=y
)
```

```
#Feature Scaling : Unlike Linear Regression, Logistic Regression benefits from scaling because it improves optimization and car
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)
```

Logistic Regression Model

A Logistic Regression classifier was trained using the standardized predictor variables to estimate the probability that a passenger is satisfied.

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(
    max_iter=1000,
    random_state=42
)

model.fit(
    X_train_scaled,
    y_train
)
```

```
#Predictions
y_pred = model.predict(X_test_scaled)

y_prob = model.predict_proba(X_test_scaled)[:,1]
```

Model Evaluation

The performance of the Logistic Regression model was evaluated using multiple classification metrics, including the Confusion Matrix, Accuracy, Precision, Recall, F1-score, and Classification Report.

```
#Acuracy
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", round(accuracy,4))
```

```
Accuracy: 0.8289
```

```
#Precision
from sklearn.metrics import precision_score

precision = precision_score(y_test, y_pred)

print("Precision:", round(precision,4))
```

```
Precision: 0.8441
```

```
#Recall
from sklearn.metrics import recall_score

recall = recall_score(y_test, y_pred)

print("Recall:", round(recall,4))
```

```
Recall: 0.843
```

```
#F1 score
from sklearn.metrics import f1_score

f1 = f1_score(y_test, y_pred)

print("F1 Score:", round(f1,4))
```

```
F1 Score: 0.8436
```

```
#Confusion Matrix
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)

cm
```

```
array([[ 9546,  2213],
       [ 2232, 11985]])
```

```
plt.figure(figsize=(6,5))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues'
)

plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")

plt.show()
```

Interpretation

The Confusion Matrix summarizes the model's classification performance by comparing actual passenger satisfaction with predicted satisfaction.

High values along the main diagonal indicate that the model correctly classified most passengers.

```
#Classification Report
from sklearn.metrics import classification_report

print(classification_report(y_test,y_pred))
```

```
              precision    recall  f1-score   support

           0       0.81      0.81      0.81     11759
           1       0.84      0.84      0.84     14217

    accuracy                           0.83     25976
   macro avg       0.83      0.83      0.83     25976
weighted avg       0.83      0.83      0.83     25976
```

Interpretation

The Classification Report summarizes Precision, Recall, F1-score, and Support for each class.

These metrics provide a more comprehensive evaluation than accuracy alone, especially when assessing the model's ability to correctly identify satisfied and dissatisfied passengers.

```
#ROC curve
from sklearn.metrics import roc_curve
from sklearn.metrics import roc_auc_score

fpr, tpr, thresholds = roc_curve(y_test,y_prob)

auc = roc_auc_score(y_test,y_prob)
```

```
plt.figure(figsize=(7,6))

plt.plot(
    fpr,
    tpr,
    label=f"AUC = {auc:.3f}"
)

plt.plot(
    [0,1],
    [0,1],
    linestyle="--"
)

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")

plt.legend()

plt.show()

print("AUC:",auc)
```

Interpretation

The ROC curve evaluates the model's ability to distinguish between satisfied and dissatisfied passengers across different classification thresholds.

An Area Under the Curve (AUC) close to 1 indicates excellent classification performance.

Feature Importance

```
#Logistic Regression coefficent
coefficients = pd.DataFrame({
    'Feature':X.columns,
    'Coefficient':model.coef_[0]
})

coefficients = coefficients.sort_values(
    by='Coefficient',
    ascending=False
)

coefficients.head(10)
```

```
plt.figure(figsize=(10,8))

sns.barplot(
    data=coefficients,
    x='Coefficient',
    y='Feature'
)

plt.title("Feature Importance (Logistic Regression Coefficients)")

plt.show()
```

Interpretation

Positive coefficients increase the probability that a passenger is classified as satisfied, whereas negative coefficients decrease that probability.

The variables with the largest positive coefficients are the strongest predictors of customer satisfaction.

Model Performance

The Logistic Regression model demonstrated strong classification performance on the test dataset.

Metric	Value
Accuracy	**82.89%**
Precision	**84.41%**
Recall	**84.30%**
F1 Score	**84.36%**
AUC	**90.35%**

Interpretation

The model achieved an accuracy of **82.89%**, indicating that it correctly classified approximately 83 out of every 100 passengers.

The precision (84.41%) indicates that when the model predicts a passenger is satisfied, the prediction is correct most of the time.

The recall (84.30%) demonstrates that the model successfully identifies the majority of satisfied passengers.

The F1 Score confirms a strong balance between precision and recall.

Finally, the AUC of **0.9035** indicates excellent discrimination between satisfied and dissatisfied passengers.

Interpretation

The confusion matrix shows that the Logistic Regression model correctly classified a large proportion of passengers.

- True Negatives: **9,546**
- False Positives: **2,213**
- False Negatives: **2,232**
- True Positives: **11,985**

Most observations fall along the main diagonal, demonstrating that the classifier performs well in distinguishing satisfied from dissatisfied passengers.

Feature Importance Interpretation

The Logistic Regression coefficients identify the variables that most strongly influence passenger satisfaction.

Strongest Positive Predictors

- Inflight entertainment
- Seat comfort
- On-board service
- Check-in service
- Ease of online booking
- Leg room service

Higher ratings for these services increase the likelihood that a passenger will be satisfied.

Strongest Negative Predictors

- Disloyal Customer
- Economy Class
- Personal Travel
- Departure/Arrival Time Convenience
- Food and Drink
- Arrival Delay in Minutes

These variables reduce the probability of passenger satisfaction, indicating potential areas for operational improvement.

Business Recommendation

The Logistic Regression analysis identified several factors that significantly influence passenger satisfaction.

Based on the model findings, the airline should prioritize:

- Improving inflight entertainment quality.
- Enhancing seat comfort across all cabin classes.
- Providing higher-quality onboard service.
- Streamlining the online booking process.
- Improving check-in efficiency.
- Enhancing leg room where operationally feasible.

The analysis also indicates that dissatisfaction is more common among Economy Class passengers, disloyal customers, and passengers traveling for personal purposes. Improving the travel experience for these customer groups may increase overall satisfaction and encourage repeat business.

Furthermore, minimizing arrival delays and maintaining consistent service quality can further improve customer perceptions of the airline.

Conclusion

This project successfully developed a Logistic Regression model to predict airline passenger satisfaction using demographic information, travel characteristics, and service quality ratings.

The dataset was cleaned, categorical variables were encoded, numerical features were standardized, and the model was evaluated using multiple classification metrics.

The final model achieved an accuracy of **82.89%**, a precision of **84.41%**, a recall of **84.30%**, an F1 Score of **84.36%**, and an AUC of **90.35%**, demonstrating strong predictive performance.

Feature importance analysis revealed that inflight entertainment, seat comfort, onboard service, and online booking experience are the strongest positive drivers of customer satisfaction, while economy class, disloyal customers, and travel delays reduce passenger satisfaction.

Overall, the model provides valuable insights that can support evidence-based decision-making aimed at improving customer experience and increasing passenger satisfaction.

```
Start coding or generate with AI.
```